

“Lightning has a very specific waveform, it is very important to know its characteristics to design circuit protection”

We are surrounded by devices containing circuits all around us in our day-to-day life, and all these circuits require protection at various levels to avoid the device from getting damaged. Many non-protected circuits have had adverse effects on human lives, and so providing protection to a circuit has become an important requirement.

Dr Bharat Sheno, global technical director of Littelfuse, talks to Dilin Anand and Sneha Ambastha about circuit protection



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What are the key things to be considered while selecting a circuit protection component for a design?

Couple of key things that the designers should know include the basic electrical parameters where the device will operate (maximum system voltage, maximum running current going through the device, etc), the maximum operating temperature as most of our products are temperature sensitive, the environment in which the components are going to work and the environmental faults the products need protection from. In order to design the correct component, we need to know faults specifically. Sometimes customers do not know that and we may over-design, which can cause higher cost and more space, and also under-design, which can cause failure. Customers also need to know how sensitive the circuit is, which means that when would it get damaged and the factors that can damage it. Let us take the example of fuses, which are not the same always. Some fuses are such that if we pass 20A through them, they would trip in one second whereas others are such that if we pass 20A through them, they would trip in one microsecond. So if a product (used for circuit protection) gets damaged in a microsecond or millisecond and cannot withstand even a second fault, then it can lead to some damage to the circuit.

What are the parameters based on which circuit protection components are selected for lightning protection?

There are fuses that protect against overcurrent and varistors that protect against overvoltage. Other than those electrical parameters, there are several environment-related parameters like lightning. Lightning has a very specific

waveform, and it is very important to know its characteristics like the peak voltage, the maximum current, the time duration and the number of pulses. IEEE and UL have developed some standards that simulate lightning—not direct lightning but the effect of lightning storms either inside or outside the building.

What are the technologies available to designers to select from, for lightning protection of electronic systems?

When you look at lightning protection or protection against overvoltage surge or overcurrent surge, there are several technologies that can be used. One of those is varistor—it is a ceramic-based technology, and we call them metal oxide varistors because they are made of zinc oxide. They are very strong and can handle a lot of surge but are very bulky. They are cost-effective (a very low-cost technology) and have been in use for many years in spite of being somewhat slow (do not react instantaneously). On the other hand, lightning takes a few microseconds, so till the time varistor starts working, some part of the lightning already passes through. A superfast technology that causes lightning protection is called TVS diode. As soon as lightning comes in, it starts working, so there is very little energy that is let through. However, it cannot handle higher surges. A varistor can handle ten times higher surges than the diodes. The diodes are very expensive, especially when they are big, because of our requirement that they should handle high surges. Then there is another technology called the gas tube. This has very high lightning protection capability but is the slowest of all. It is a technology where there is an arc within the gas tube.